

RF Smith Chart and matching problems

1. Assume you are working at RF and 50Ω plot these values on the smith chart:
 - a. 200Ω
 - b. 50Ω
 - c. 100Ω
 - d. 15Ω
 - e. 0Ω
 - f. 2500Ω
 - g. A short cct
 - h. An open cct
 - i. Change each of these values to admittances using the smith chart, read off the values and state what the component values of admittance would be for these.
2. Plot b, c and d again but this time assume you are working on a 75Ω system, what value places you in the centre of the chart.
3. If you were working with a usb receiver with transmission lines with $Z_0 = 90\Omega$ what normalized value would give you 180Ω
4. In a 50Ω system plot the following reactances:
 - a. $j200\Omega$
 - b. $j50\Omega$
 - c. $j15\Omega$
 - d. $j5\Omega$
 - e. $-j20$
 - f. $-j75$
 - g. $-j250$
5. In a 50Ω system operating at 200MHz there is a 3.9pF capacitor. Plot its position on the smith chart if:
 - a. It were in series
 - b. In parallel
6. What would be the equivalent value and connection orientation (series/parallel) of an inductor for each of the points plotted in 5.
7. Repeat questions 5 and 6 but at a frequency of 1GHz.
8. A 50Ω transmission line is terminated with a complex load with the value $70+j50\Omega$. Using the smith chart find:
 - a. The reflection coefficient Γ
 - b. The voltage standing wave ratio in dB and as a ratio
 - c. Plot on the smith chart all values that would provide the same SWR as b
 - d. At 150MHz what series values of resistance and reactance would provide this load.
9. What is the conjugate value of the load used in 8?